1. Abstract

In this project, I used machine learning to see if we could predict heart disease using a mix of clinical and physiological data. I took a heart disease dataset, cleaned it up by turning categories into factors, and split it into training and testing groups. I tested three specific models: a decision tree, logistic regression, and k-nearest neighbors (KNN). To see how they performed, I looked at their accuracy, sensitivity, and specificity. Logistic regression and KNN were the most accurate overall, but KNN had the best sensitivity. That makes it the most useful tool here for catching actual cases of heart disease. These results show that machine learning can be a real help in making clinical decisions.

2. Introduction

Heart disease is a massive issue globally. If we can catch it early, we can get people the medical help they need before things get worse. I wanted to see if machine learning models could take clinical and physiological data and accurately tell us whether a patient has heart disease or not.

My main hypothesis for this study is that there is a clear link between these clinical features and heart disease classification. I am betting that machine learning models can pick up on those patterns and accurately predict when heart disease is present.

3. Methods

I started with a heart disease prediction dataset to try and classify patients. The main goal was to predict the "HeartDisease" variable, where a 0 meant the patient was healthy and a 1 meant they had heart disease.

Once the data was loaded, I spent some time cleaning and preparing it. I had to convert several categorical variables into factors so the models could handle them. This included things like sex (male or female), the type of chest pain they had, their resting ECG results, whether they got angina during exercise, and the category of their ST slope. For chest pain, the data looked at typical angina (TA), atypical angina (ATA), non-anginal pain (NAP), and those who were asymptomatic (ASY). The ECG results were grouped into normal, ST-T wave abnormalities (ST), and left ventricular hypertrophy (LVH). Exercise-induced angina is basically chest pain that hits during physical activity when blood flow is restricted, and the ST slope tracks the pattern of the ST segment on an EKG as Up, Flat, or Down.

I also scanned the dataset for any missing values or duplicate entries. I found some duplicate rows and removed them to keep the results honest. After that, I split the data into two groups: 70% for training the models and 30% for testing them. The training set is where the models learned the patterns, while the testing set let me see how they performed on data they had not seen yet.

I ended up implementing three different machine learning models: a decision tree, logistic regression, and k-nearest neighbors (KNN). I used the decision tree to classify the heart disease cases and to figure out which variables were the most important for making a prediction. I used logistic regression to get a probability of heart disease for each patient, and KNN worked by looking at how similar a specific data point was to the ones around it.

To judge how well these models worked, I focused on accuracy, sensitivity, and specificity. Sensitivity was the big one for me because it tells you how good the model is at finding the people who are sick.

4. Results

After I ran the code, the numbers came in. The decision tree hit an accuracy of **82.97%**, with a sensitivity of **83.87%** and a specificity of **81.82%.** Logistic regression did a bit better, reaching **87.68%** accuracy, **89.03%** sensitivity, and **85.95%** specificity. The k-nearest neighbors model also hit that **87.68%** accuracy mark, but it had the highest sensitivity of the bunch at **90.97%**, along with a specificity of **83.47%.**

Looking at these results, it is clear that logistic regression and KNN were the top performers for accuracy. KNN is the winner if you care most about sensitivity—it is the best at flagging patients who have heart disease. On the other hand, logistic regression had the highest specificity, so it was slightly better at correctly identifying healthy patients. The decision tree also gave some insight into the "why," pointing to ST slope as the most important predictor, followed by chest pain type, the ST peak, and exercise-induced angina.

5. Discussion

These results back up my hypothesis. The machine learning models were able to predict heart disease with a high degree of accuracy. While logistic regression and KNN were the strongest overall, the decision tree still did a decent job, even if its performance was a little lower.

I put a lot of weight on sensitivity in this project. A false negative is a scary outcome here—it means a patient with heart disease gets told they are fine. Since KNN had the highest sensitivity, it is the best model for making sure we do not miss those cases. Logistic regression was also a strong contender, and because it had the highest specificity, it was the most reliable at identifying the patients who did not have heart disease.

The decision tree was still useful because it is easy to interpret. Seeing that ST slope was the top predictor suggests that the changes we see on an ECG during exercise are a huge red flag for heart disease. It also highlighted chest pain and exercise-induced angina, which makes total sense since those are the classic symptoms doctors look for anyway.

6. Conclusion

Overall, these models did a great job predicting heart disease status. Logistic regression and KNN were the heavy hitters, with KNN being the go-to for catching as many cases as possible through high sensitivity. The decision tree might have been less accurate, but it was great for showing which clinical features matter. Moving forward, I think we should focus on pushing that sensitivity even higher to minimize false negatives and make sure no one slips through the cracks.

7. References

fedesoriano. (September 2021). Heart Failure Prediction Dataset. Retrieved [04/25/2026] https://www.kaggle.com/fedesoriano/heart-failure-prediction.